AquaLink

Connecting South African Villages to Clean, Reliable Water

AQUALINK SOLUTION ARCHITECTURE

Smart Water Access. Reliable Supply. Stronger Communities.

1. PRESENTATION LAYER
(User Interfaces)

1.1 Household Portal (Subscriber)
- Register and log in
- Choose a water subscription plan
- View water usage
- View payment status and payment information
- Receive service and system alerts

1.2 Supplier Dashboard (Borehole Owner)
- Manage the registered borehole
- Monitor water flow
- View tank levels
- Track income and payments
- Receive alerts relating to water supply and the system

1.3 Administrator Dashboard
- Manage households and subscribers
- Manage suppliers and boreholes
- Manage subscriptions
- Monitor water usage
- Track payments
- Manage system alerts and reports

2. APPLICATION / API LAYER
(Business Logic & Services)

Application Modules

User Management
Registration, login, roles and user information.

Household Management
Household records, addresses, household size and user association.

Supplier Management
Supplier records and borehole ownership information.

Subscription Management
Water plans, subscription status and monthly fees.

Payment Management
Payment records and payment status.

Water Usage Management
Processing and storing water usage information from sensors.

Alert Management
Generating and managing water/system alerts.

Reporting & Analytics
Providing usage, payment and operational reports.

Core Services

- RESTful API Gateway
- Authentication & Authorization
- Business Logic Layer
- File / Document Service
- Background Jobs / Scheduler

3. DATA LAYER
(Data Storage)

3.1 Database (Relational)

Entity	Main Information
Users	user_id, name, email, role
Households	household_id, address, size, user_id
Suppliers	supplier_id, name, contact, borehole_id
Boreholes	borehole_id, supplier_id, location
Subscriptions	subscription_id, household_id, plan, status, monthly_fee
Payments	payment_id, household_id, amount, date, status
Sensor Readings	reading_id, sensor_id, litres_supplied, timestamp, water_level
Alerts	alert_id, sensor_id, type, message, status

3.2 External Services

Payment Gateway
Supports payment processing such as mobile money or card payments.

Notification Service
Supports SMS, email or push notifications.

Internet / IoT Network
Provides communication between connected sensors and the platform.

4. PHYSICAL / IoT LAYER
(Water Infrastructure & Sensors)



1. **Borehole**
Water is pumped from the borehole.

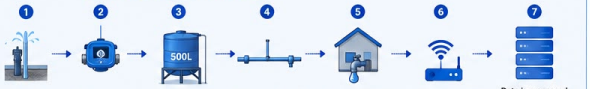
2. **Flow Sensor**
A digital flow meter records water supplied in real time.

3. **500L Buffer Tank**
A gravity-fed buffer tank smooths supply peaks and stores water.

4. **HDPE Pipe Network**
20mm HDPE pipes run along the street and branch to registered households.

5. **Household Tap**
Each subscriber has a dedicated household tap.

5. END-TO-END SYSTEM FLOW



1. Borehole supplies water.
2. Flow sensor measures the water supplied.
3. 500L buffer tank stores water and smooths demand.
4. HDPE pipe network distributes water to registered households.
5. Households access water through their dedicated taps.
6. Sensor information is transmitted to the AquaLink platform.
7. Data is processed, stored and made available to users (Dashboards, Alerts, Reports, Payments).

6. KEY BENEFITS

Reliable Water Access
Provides continuous and monitored water supply.

Real-time Monitoring
Tracks usage, tank levels and potential issues.

Transparency
Clear payments, reports and supplier tracking.

Smart Alerts
Immediate onfications for leaks, low tank levels and system issues.

Community Empowerment
Creates jobs, supports local suppliers and strengthens communities.

Scalable & Sustainable
Designed to expand to more villages and communities.

TECHNOLOGY STACK (EXAMPLE)

- Frontend: React / Next.js (or similar)
- Backend: Node.js / Django (or similar)
- Database: PostgreSQL
- IoT Protocols: MQTT / HTTP
- Cloud / Hosting: AWS / Azure (or similar)
- Payments: Payfast / Yoco / Mobile Money
- Notifications: Twilio / SMS / Email

AquaLink – Smart Technology. Clean Water. Stronger Communities.